Speech production: Impact of formants on produced speech

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2026

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1 Introduction

1.1 Academic Context

This project was made for the validation of the Prosody course that took place during the second semester of the first year of the NLP master at IDMC, Nancy.

This part of this course was supervised by Mélanie Lancien. All the Praat scripts used in this project are under her intellectual property.

A GitHub repository can be found here: https://github.com/MaeDj/M1_NLP_S2_Prosody_Lancien
You will find on it all the resources including raw CSV files with all the computed values and used algorithms.

1.2 Purpose

The purpose of this project is to study the formants of a given audio file and compute feature metrics from it.

- F0 curve
- Mean (F0,F1,F2,F3)
- Median (F0,F1,F2,F3)
- Standard deviation (F0,F1,F2,F3)
- F1-F3 dimension schema

2 Data Description

The chosen base audio is "The-Very-Hungry-Caterpillar.wav". This audio is 2min18 long and presents a clear spectrogram without parasitic noises, clear formants and a balanced level of energy.

Audio features were extracted via Praat and the "fixed_formants_extraction.praat" script, an adapted version of "ScriptProsody.praat" originally made by Mélanie Lancien and given as the base script for this project.

This script provided .txt output (resultThe-Very-Hungry-Caterpillar.txt) that were converted into a .csv file (The-Very-Hungry-Caterpillar.csv) to simplify the next computation step.

2.1 Formants definition

Formants can be observed for vowels. They represent voice frequency, in Hz, obtained by the vibration of the vocal cords.

2.1.1 F0

F0 is a specific formant. It is not directly extracted from the speech signal but computed from it. The goal of computing F0 is to understand the speech signal globally, approximating it to a "(quasi-)periodic structure of voiced speech signals" and deducing its frequency.

2.1.2 F1

F1 is the first formant (the lower one on the spectrogram, the nearest to the voiced bar). Its main cause is the lowering of the jaw and tongue during speech production. It is the one with the lowest frequency (around 500 Hz), which explains why it is often correlated with high energy because there is more empty space in the mouth.

2.1.3 F2

F2 is the second formant (the second one after the nearest to the voiced bar). It is linked to backward movements of the tongue and lip rounding.
When lips are not rounded, it is possible to deduce vowel backness from it by making the difference between F1 and F2.

2.1.4 F3

F3 is the third formant (the one above F2). It is not the last formant but these three are the most used and explainable. It is linked to the roundness of the vowel in general.

3 Formant processing

3.1 F0 curve

The F0 curve is an important tool for visualizing global speech behavior. It helps to better understand the speaker's prosody and deduce some speech particularities. Of course, unvoiced phonemes do not have an F0 curve, but this representation can still be useful at a sentence level.

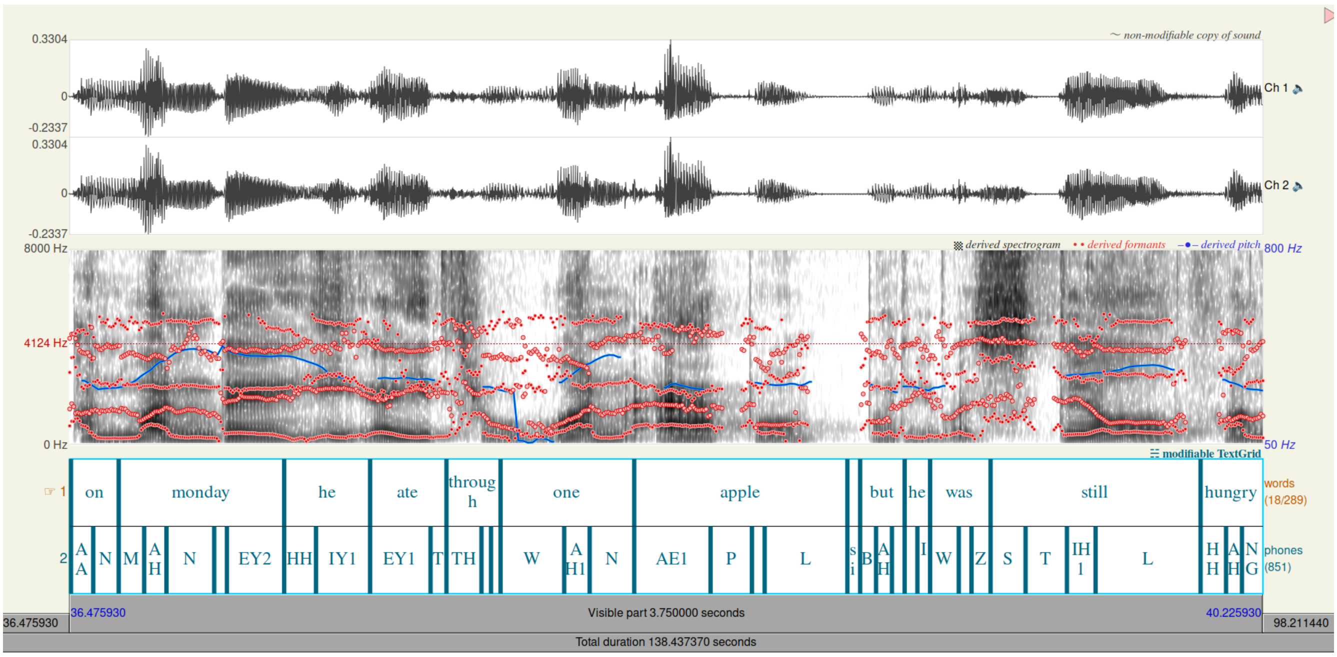

Figure 1: F0 curve (in blue) from 36.5s to 40.2s

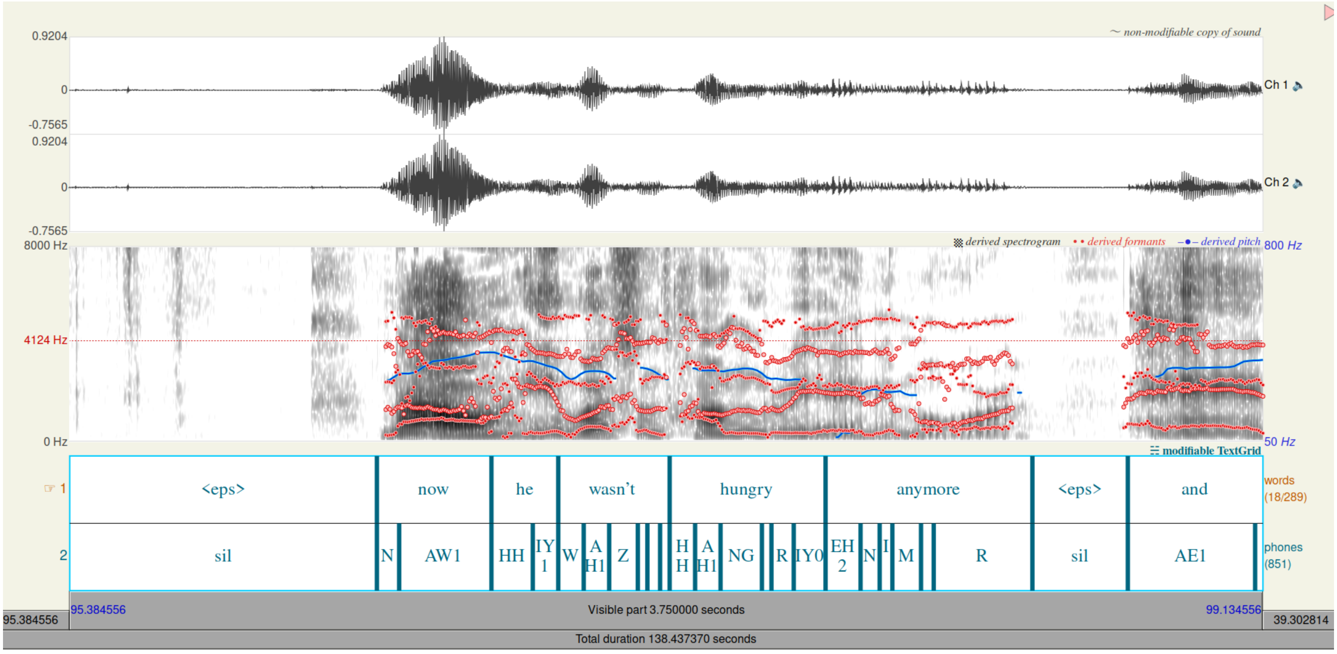

Figure 2: F0 curve (in blue) from 85.4s to 99.13s

3.1.1 Results

The F0 curve clearly represents reading behavior with a drop in pitch at the end of each sentence and few F0 peaks linked to the non-spontaneous and low-emotional speech.

We can also see that the intonation marks an emphasis provoked by the accent of the speaker and the accentuation on some phonemes to keep the listener engaged. ("On Monday", 1)

3.2 Formants across the global audio

[1]

3.2.1 F0

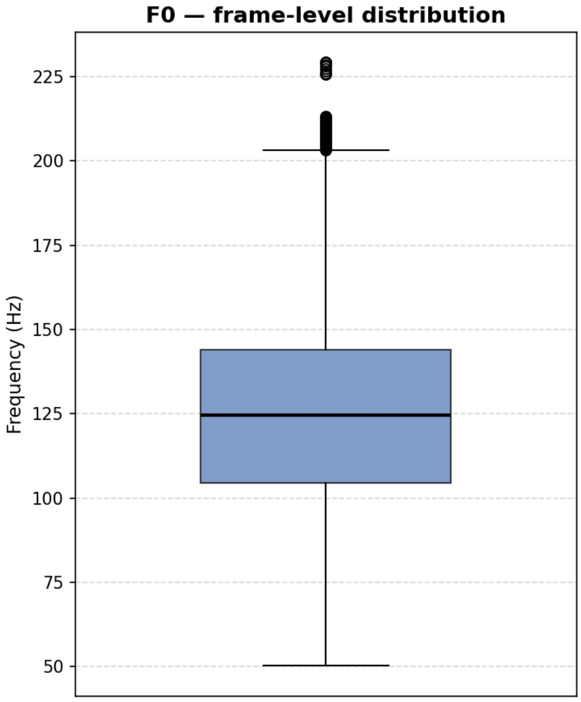

Figure 3: Boxplot representation of F0 metrics across the global audio

[1] Please find the detailed CSV files with all the data on: https://github.com/MaeDj/M1_NLP_S2_Prosod_Lancien

3.2.2 F1

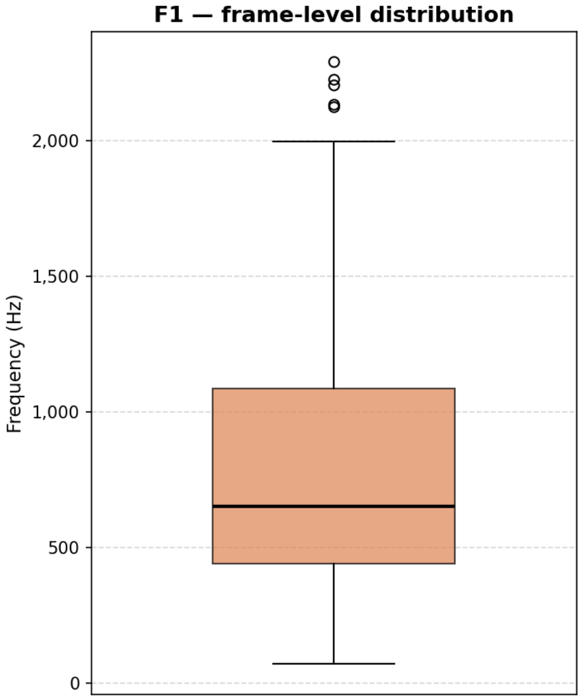

Figure 4: Boxplot representation of F1 metrics across the global audio

3.2.3 F2

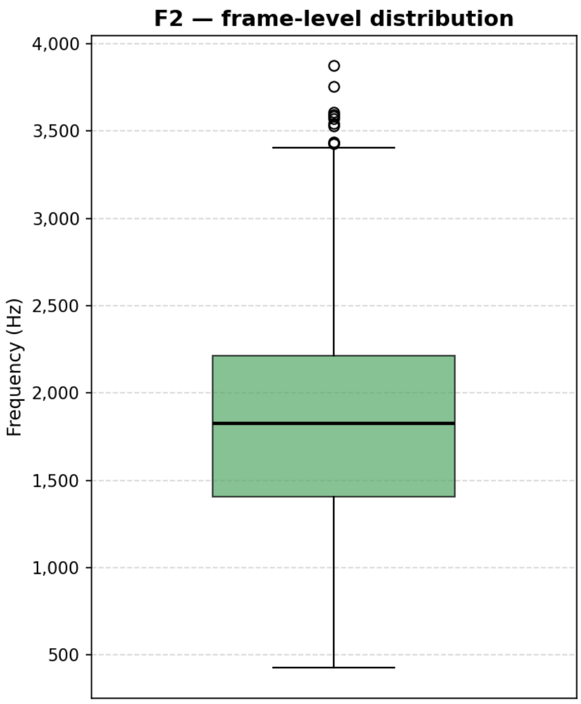

Figure 5: Boxplot representation of F2 metrics across the global audio

3.2.4 F3

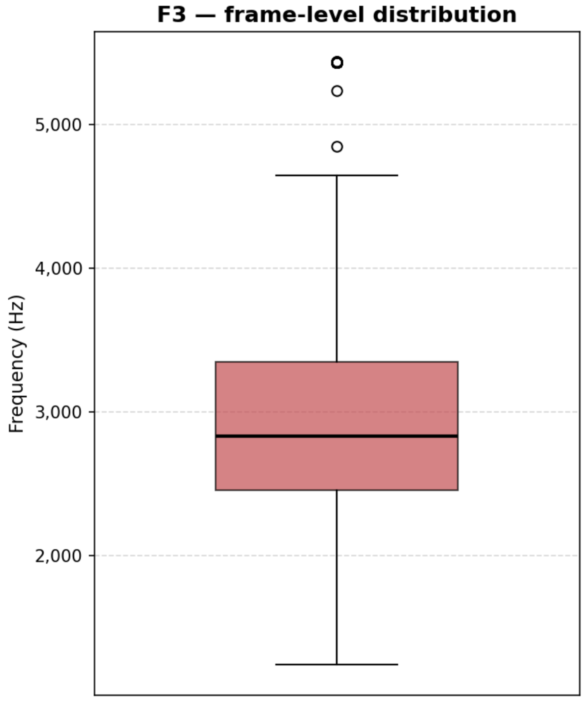

Figure 6: Boxplot representation of F3 metrics across the global audio

3.2.5 Results

The study of statistical metrics on different formants also brings information about speech production. The F0 median and quantile represented in figure 3 clearly show the speaker has a male vocal tract (F0 between 90 and 155Hz) but does not have a particularly deep voice, which could indicate a smaller vocal tract than a person with a lower F0. Moreover, this person presents a great F0 variance that can be explained by the type of speech task. “The Hungry Caterpillar” is a children’s story. Storytellers of this kind of story often have to exaggerate their intonation to get the audience’s attention.

F1, F2 and F3 are less relevant at a global level because they usually show very precise production phenomena.
But F1 value distribution confirms what was said before about intonations. The speaker’s F1 presents a high maximum representing a momentary concentration of energy and an exaggerated lowering of the jaw and tongue explained by the type of task. This phenomenon can also be seen for F2 and F3, showing deep modifications of the vocal tract during speech.

3.3 Formants across sentences of the audio

[2]

3.3.1 Mean

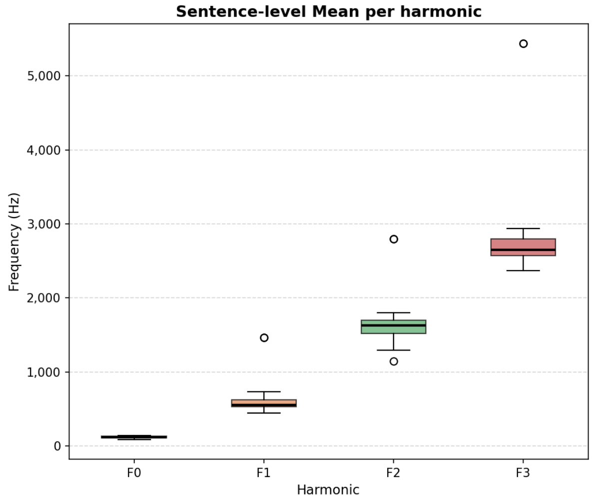

Figure 7: Boxplot of mean distribution of each formant across all audio sentences

3.3.2 Median

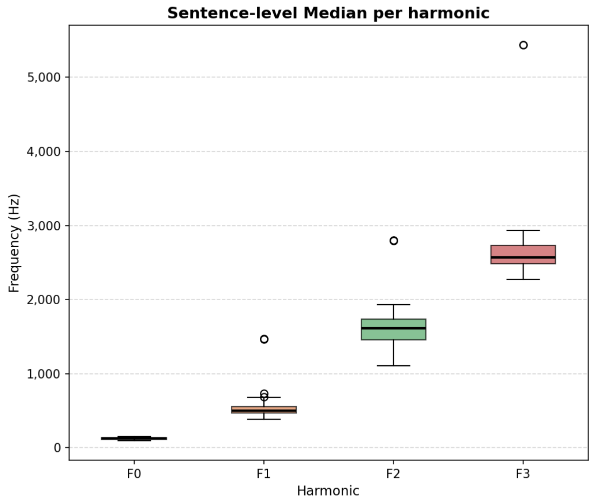

Figure 8: Boxplot of median distribution of each formant across all audio sentences

[2] Please find the detailed CSV files with all the data on: https://github.com/MaeDj/M1_NLP_S2_Prosod_Lancien

3.3.3 Standard Deviation

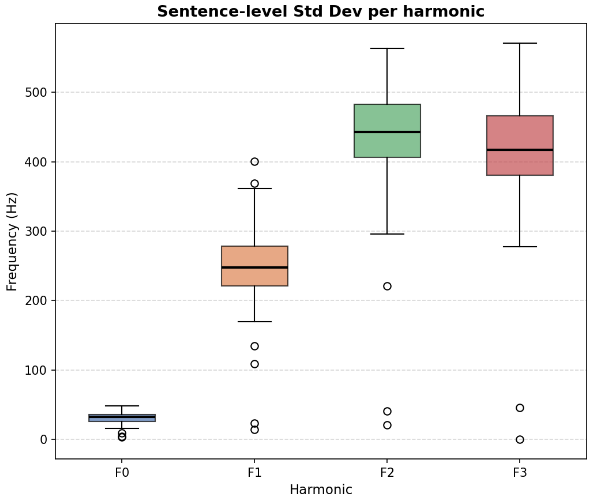

Figure 9: Boxplot of standard deviation distribution of each formant across all audio sentences

3.3.4 Results

Sentence-level analysis brings additional information about the speaker's speech production. We still observe deep speech variation between sentences themselves as shown in 9, while the median 8 and the mean 7 remain consistent, showing speech stability despite the variation.
This can be attributed to the control of speech, differentiating the speech production of someone speaking excitedly to another person and, as here, someone telling a story and trying to make it both understandable and interesting to hear.

3.4 F1-F3 dimension schema

The F1-F3 three-dimensional schema is useful for representing a global and visual overview of the most commonly used phonemes, as well as particular articulation patterns in speech production behavior.

Here, for practical reasons, a two-dimensional representation was chosen. This representation is not optimal for observing global speech production behavior but is more efficient for identifying specific speech particularities arising from specific formants.

A linear regression was also represented (in black) on each graph. This metric is more of an indicator of the global behavior of one formant compared to another, but is less meaningful in terms of speech behavior compared to a global analysis of the schema.

3.4.1 F1-F2 dimension schema

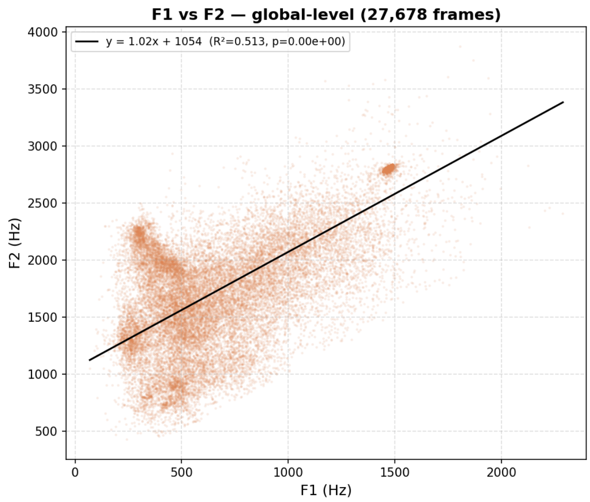

Figure 10: Two-dimensional schema: F1 against F2

3.4.2 F1-F3 dimension schema

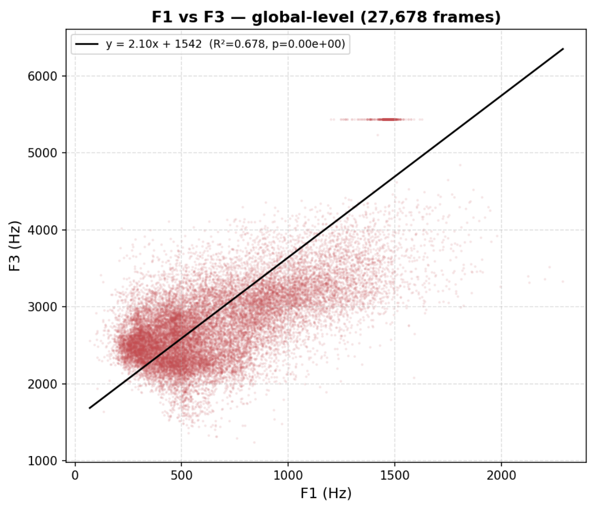

Figure 11: Two-dimensional schema: F1 against F3

3.4.3 F3-F2 dimension schema

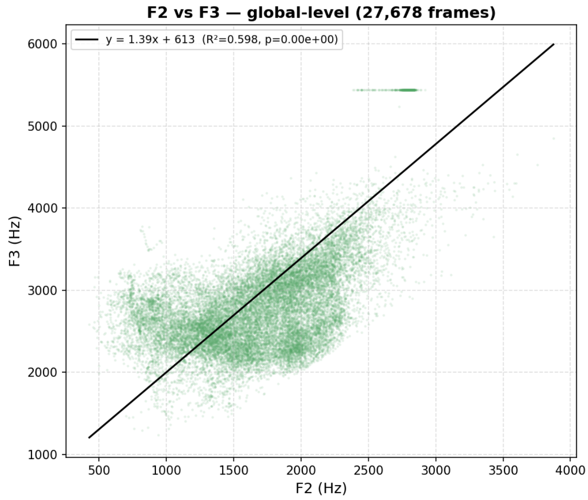

Figure 12: Two-dimensional schema: F2 against F3

3.4.4 Results

The F1-F3 dimension schema shows the speaker's tendency to round (dot cloud homogeneous/concentrated on the higher values of F2 and F3 12) and to open the mouth widely (dot cloud concentrated on the lower values of F1 and F2 10).

In terms of speech production, this metric behavior perfectly fits our previous analysis regarding the type of speech task and the speaker's behavior and particularities.

4 Limitations

Speech processing is a very demanding task requiring a great deal of rigor and technical skill. Any process applied to voice data can easily lead to mistakes that cannot be corrected, because feature extraction encodes the signal in some way, making it impossible to re-check.
Because of this particularity, making inferences from speech analysis can be very difficult and requires a great deal of work, verification and methodological care to ensure correct processing and avoid drawing wrong conclusions based on biased data.

5 Conclusion

Speech processing and formant analysis are powerful tools for better understanding speech production.
These techniques can be very useful and powerful in multiple tasks but require deep interdisciplinary understanding of both linguistics and computer science, as well as an awareness of their respective limitations.
Considering these facts, creating guidelines and protocols is a genuine challenge that could become a key turning point in the NLP field.